

Scattering angle dependence of Fano resonance profiles in cold atomic collisions analyzed with the complex valued w parameter

T Singh^{1,5}, R A Singh^{1,6}, F Koike^{2*} , M Iizawa^{3,7} and Y Azuma^{2,4*}

¹Department of Physics, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India

²Department of Materials and Life Sciences, Sophia University, Tokyo 102-8554, Japan

³Department of Physics, Rikkyo University, Tokyo 171-8501, Japan

⁴Department of Physics, Birla Institute of Technology and Science-Pilani, K. K. Birla Goa Campus, Goa 403726, India

⁵ Present Address: School of Earth and Space Exploration, Arizona State University, Tempe, AZ 85281, USA

⁶Centre for Astrophysics and Supercomputing (CAS), Swinburne University of Technology, Hawthorn, Australia

⁷Institut für Theoretische Physik, Technische Universität Braunschweig, Mendelssohnstr. 3, D-38106 Braunschweig, Germany

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Abstract: The scattering angle dependence of Fano resonance profiles in cold atomic collisions has been theoretically studied. A complex-valued parameter w with an analytical formula describing the asymmetry of the resonance profile is proposed as a new development from previous work on electron resonance scattering from atoms [F. Koike, J. Phys. **B10**, 2883 (1977)]. It serves as the general formulation which leads to the angle-dependence of Fano's q parameter. Calculations for cold elastic-collisions of hydrogen atoms with krypton atoms have been carried out. Strong angle-dependence is demonstrated for the resonance profile asymmetry in the differential scattering cross section due to the interference with non-resonant partial waves. The scattering angle dependence of the resonance profile and thus the newly proposed asymmetry parameter is highly sensitive to the inter-atomic interaction potentials. This is likely to prove useful for the study of interaction potentials themselves.

Keywords: Cold atomic collisions; Fano resonance; Angle dependent profile

1. Introduction

Fano resonances, characterized by Fano's q parameter, have been extensively studied since their inception [1]. The concept was originally studied in atomic and molecular physics but subsequently found use in various phenomena including classical mechanics, electrical engineering, modern optics etc. [2]. Three tiers can be recognized for applying the concept in quantum mechanical resonance scattering depending on the energies and physical scale. They are 1. nuclear collisions, 2. electron scattering and photoionization from atoms and molecules, and 3. the recently evolving area of cold atomic and molecular

collisions. Wigner [3, 4] discussed the nuclear reactions and proposed Wigner's one-level resonance formula. Gibson and Dolder [5] observed the angle-dependent asymmetry of the resonance profiles in the differential elastic electron scattering from helium atoms. Koike [6] pointed out that this asymmetry can be described by an angle-dependent complex valued w parameter, and also that w produces the angular dependence of the Fano's q parameter.

In general, a well-isolated quantum mechanical resonance can be observed only when the de Broglie wavelength of the collision system is comparable to the range of interaction between the colliding particles. For atomic and molecular collisions, it is the "cold collision" that satisfies such a condition. The incident particle may be temporarily trapped when the system undergoes resonance scattering. Such "orbiting resonances", characterized by the formation of a quasi-bound state of the colliding partners, were observed by Toennies et al. [7] in collisions of

*Corresponding author, E-mail: koikef@sophia.ac.jp; y-azuma@sophia.ac.jp

hydrogen atoms and hydrogen molecules with rare gas atoms at the center-of-mass collision energies ranging from 0.5 to 100 meV.

Subsequently, such resonance phenomena in “cold” atomic and molecular collisions have been studied by a number of authors [8–18]. Recently, the scattering angle dependence of the resonance profiles for cold $\text{He}^* + \text{D}_2$ collisions was experimentally measured at the energy of 4.8 Kelvin by Paliwal et al. [17]. They discussed the asymmetry of the resonance profiles that appeared in the energy spectra of the differential scattering cross-sections in terms of Fano resonance [1]. However, as discussed later in Sect. 4, their data analysis failed to avoid inadequacies from the perspective of a theoretical description of the resonance profile. Consequently, it is crucial to establish the proper methodology for the resonance spectral analysis in cold atomic and molecular collisions. The angle dependence of the asymmetry in the resonance profiles gives a very sensitive tool to determine the inter-atomic interaction potential which leads to van der Waals forces. In the present paper, we consider the angle-dependent Fano-type asymmetry of the resonance profiles in cold atomic and molecular collisions following the derivation given in ref [6].

Quantum mechanical resonances take place when a scattering wavefunction is pulled into the interaction region to form a quasi-bound state. The scattering phaseshift changes by π radian when the collision energy passes through the resonance point. The asymmetry of the profiles for an isolated resonance in elastic scattering is determined solely by the background phaseshifts of the collision system [6]. In the present paper, we shall apply the formalism developed for electron scattering [6] to the resonance in cold atomic and molecular collisions. An angle-dependent resonance profile parameter w that is analytic on the Gauss plane is introduced. A generalized mathematical expression for the angle-dependent real-valued Fano profile parameter q is given in terms of the complex-valued w parameter. The orbiting resonances in $\text{H} + \text{Kr}$ collisions are discussed in detail using the inter-atomic interaction potentials that are proposed by Toennies et al. [7]. It is shown that the w parameter describing the angle-dependent resonance profile becomes a very stringent tool to analyze the inter-atomic interaction potentials.

In the following section, we develop the theory of our complex w parameter describing angle-dependent resonance profiles in elastic differential scattering. In Sect. 3, we theoretically calculate the actual case of $\text{H} + \text{Kr}$ collisions. In Sect. 4, we discuss the characteristics of the calculated resonance profiles and develop the methodology of angle-dependent resonance profile analysis. Finally, in Sect. 5, we give concluding remarks.

2. Theory

We consider the elastic scattering of particles in which the effect of spin or the internal angular momentum of the colliding pair can be neglected. According to Blatt and Biedenharn [19], the angle-differential scattering cross section $d\sigma/d\Omega$ can be given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{k^2} \sum_{L=0}^{\infty} B_L P_L(\cos \theta) \quad (1)$$

with

$$B_L = \sum_{\ell=0}^{\infty} \sum_{\ell'=|\ell-L|}^{\ell+L} (2\ell+1)(2\ell'+1)[C(\ell, \ell' L; 00)]^2 \times \sin \eta_{\ell} \sin \eta_{\ell'} \cos(\eta_{\ell} - \eta_{\ell'}), \quad (2)$$

where k is the wavenumber, ℓ and ℓ' are angular momentum quantum numbers, η_{ℓ} and $\eta_{\ell'}$ are the elastic scattering phaseshifts of the collision system, respectively, $C(\ell \ell' L; 00)$ the Clebsch-Gordan coefficients [20], θ the center-of-mass scattering angle, and $P_L(\cos \theta)$ the Legendre polynomial of the order L . The angle-integrated cross section σ_{ℓ} for partial wave ℓ is given by

$$\sigma_{\ell} = \frac{4\pi}{k^2} (2\ell+1) \sin^2 \eta_{\ell}. \quad (3)$$

When we have an isolated resonance at $\ell = \ell_r$, we can represent the phaseshifts η_{ℓ} in terms of the resonance phaseshift Δ_{ℓ_r} and the background phaseshift δ_{ℓ} as follows.

$$\eta_{\ell} = \delta_{\ell_r} + \Delta_{\ell_r} \text{ for } \ell = \ell_r, \quad \eta_{\ell} = \delta_{\ell} \text{ for else.} \quad (4)$$

For a resonant partial wave $\ell = \ell_r$, we have $\eta_{\ell} = \eta_{\ell_r} = \delta_{\ell_r} + \Delta_{\ell_r}$ in Eq. (3). By introducing a reduced scattering energy $\epsilon = -\cot \Delta_{\ell_r}$ and a profile parameter $q = -\cot \delta_{\ell_r}$, we obtain σ_{ℓ_r} in the form of Fano profile formula [2].

$$\sigma_{\ell_r} = \frac{4\pi}{k^2} (2\ell_r+1) \frac{1}{(1+q^2)} \frac{(q+\epsilon)^2}{(1+\epsilon^2)}. \quad (5)$$

The reduced energy ϵ is related to the center-of-mass scattering energy E , resonance point energy E_r , and the resonance width Γ as $\epsilon = (E - E_r)/(\Gamma/2)$.

For the angle-differential cross sections, the resonance profile is modified by interference with partial waves having $\ell \neq \ell_r$. Following the paper by Koike [6], we can derive a resonance profile formula for $d\sigma/d\Omega$ by introducing an angle-dependent complex-number parameter w_{ℓ_r} , which gives the characteristics of the resonance profile. We take $i = \sqrt{-1}$. Then we have

$$w_{\ell_r}(\theta) = \exp(2i\delta_{\ell_r}) \sum_{L=0}^{\infty} \{(2\ell_r + 1)^2 [C(\ell_r \ell_r L; 00)]^2 + \sum_{\ell=0, \ell \neq \ell_r}^{\infty} (2\ell + 1)(2\ell_r + 1) [C(\ell \ell_r L; 00)]^2 \times [1 - \exp(-2i\delta_{\ell})]\} P_L(\cos \theta). \quad (6)$$

The cross-section $d\sigma/d\Omega$ can be given by a half of the real part of

$$F_{\ell_r} \equiv -\frac{1}{k^2} w_{\ell_r} \exp(2i\delta_{\ell_r}) + z = -\frac{1}{k^2} w_{\ell_r} \frac{\epsilon - i}{\epsilon + i} + z, \quad (7)$$

where z is the background part cross-section. The detailed derivation of Eq. (6) and the explicit expression of z are given in Appendix.

Introducing the angle-dependent Fano's q parameter

$$q(\theta) \equiv -\cot\left(\frac{1}{2} \arg(w_{\ell_r}(\theta))\right), \quad (8)$$

we have

$$\frac{d\sigma}{d\Omega} = \frac{1}{k^2} |w_{\ell_r}| \frac{1}{1 + q^2} \frac{(q + \epsilon)^2}{1 + \epsilon^2} + \frac{1}{2} (z - |w_{\ell_r}|), \quad (9)$$

where $\arg(w_{\ell_r}(\theta))$ in Eq. (8) gives the argument of the complex-number $w_{\ell_r}(\theta)$. The angle-dependent $q = q(\theta)$ reduces to the ordinary q -parameter as introduced in Eq. (5) when all the phaseshifts of non-resonant partial waves are vanishing. The parameter $q = q(\theta)$ defined in Eq. (8) is a natural extension of the ordinary angle-independent q -parameter.

We can point out here that the resonance profile in the resonant elastic differential scattering cross section can be represented by the Fano profile formula [1] with a scattering angle dependent parameter $q = q(\theta)$ reduced from the complex-number parameter $w_{\ell_r}(\theta)$ as in Eq. (8). The absolute value of w_{ℓ_r} is connected to the height of the resonance profile, and $\arg(w_{\ell_r})$ gives the asymmetry of the resonance profile. Thus, the parameter w_{ℓ_r} completely characterizes the angle dependence of the resonance profile. Furthermore, as found in Eq. (6), we may note that w_{ℓ_r} is effectively given by a polynomial of $\cos \theta$; w_{ℓ_r} is analytic with respect to $\cos \theta$ or θ .

3. Resonances in H + Kr collisions

Toennies et al. [7] studied the molecular beam scattering of hydrogen atoms/molecules from rare gas atoms, which helped in modeling the inter-atomic interaction potentials. Extensive comparison with spherically symmetric model potentials has revealed that all the observed resonances can be characterized as orbiting resonances. The resonance energy positions are highly sensitive to the potential forms.

They proposed the inter-atomic interaction potentials $V(r)$ as a function of the inter-atomic distance r of the following form.

$$V(r) = \begin{cases} D_e \left[\left(\frac{r_m}{r} \right)^{12} - 2 \left(\frac{r_m}{r} \right)^6 \right] & \text{for } r < r_s \\ -\frac{C_6}{r^6} - \frac{C_8}{r^8} - \frac{C_{10}}{r^{10}} & \text{for } r \geq r_s \end{cases}, \quad (10)$$

where D_e and r_m are the well depth and the equilibrium distance of the Lennard–Jones potential, respectively. They fitted their experimental data to the Lennard–Jones potential for the interior area of r . In the asymptotic region of r , the higher order terms of the dispersion force become significant. They introduced the dispersion force parameters C_6 , C_8 , and C_{10} for the outer area of r that is larger than the cut-off radius $r = r_s$. For further details, see Toennies et al. [7]. In Table 1, the potential parameters of H + Kr collision system are tabulated. We use this set of parameters for scattering calculations in the present paper.

In Fig. 1, we show the effective inter-atomic interaction potential of the partial waves $V(r) + \ell(\ell + 1)/(2mr^2)$ for $\ell = 0 - 6$, where m is the reduced mass of the collision system. As discussed later in this section, the $\ell = 4$ partial wave undergoes resonance at $E = 0.512 \text{ meV} = 4.13 \text{ cm}^{-1}$. The resonance point energy is indicated by thick (red online) double dot dashed horizontal line. At $\ell = 4$, the effective potential makes a slight hump around $r = 12 a_0$ (Bohr radius). The $\ell = 4$ partial wave is tunneling into the potential well to undergo resonance. In the potential energy curves for $\ell = 5$ and 6, we can also find clear humps around $r = 10 a_0$, suggesting the presence of resonance. Toennies et al. [7] have pointed out the prominent resonances at $E = 1.59 \text{ meV} = 12.8 \text{ cm}^{-1}$ for $\ell = 5$ and $E = 2.94 \text{ meV} = 23.7 \text{ cm}^{-1}$ for $\ell = 6$. In the present paper, we restrict our analysis to the $\ell = 4$ resonance.

To obtain the elastic scattering phaseshifts and cross sections, we consider the following Schrödinger equations for the ℓ -partial wavefunction $\lambda_{\ell}(r)$. We take

Table 1 The inter-atomic interaction potential of H + Kr collision system [7]. For notations r , D_e , r_m , C_6 , C_8 , C_{10} , and r_s , see Eq. (10) and their followings. The unit of length is given in a_0 (Bohr radius).

Region	Model	Parameters
$r < r_s$	Lennard–Jones	$D_e = 5.90 \text{ meV}$ $r_m = 6.75 a_0$
$r \geq r_s$	$-\frac{C_6}{r^6} - \frac{C_8}{r^8} - \frac{C_{10}}{r^{10}}$	$C_6 = 7.80 \times 10^5 \text{ meV} a_0^6$ $C_8 = 1.57 \times 10^7 \text{ meV} a_0^8$ $C_{10} = 4.05 \times 10^8 \text{ meV} a_0^{10}$
$(r_s = 9.13 a_0)$		

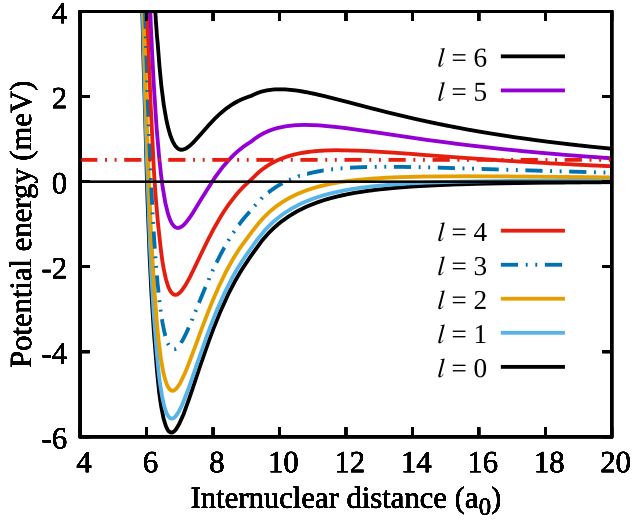


Fig. 1 Effective inter-atomic interaction potential $V(r) + \ell(\ell+1)/(2mr^2)$ for $\ell=0-6$ are shown in order of increasing potential energy. Solid(black online) curve at the bottom is of $V(r)$ given in Eq. (10). Solid(red online) curve at the fifth from the bottom is of the $\ell=4$ partial wave, which undergoes the shape resonance. Horizontal double dot dashed(red online) line at $0.512 \text{ meV} = 4.13 \text{ cm}^{-1}$ indicates the resonance point energy of $\ell=4$ resonant partial wave. Potential energy is given in units of meV. Inter-atomic distance is given in units of a_0 (Bohr radius)

$U(r) = 2mV(r)$. We use atomic units $e = \hbar = m_e = 1$ throughout the paper unless otherwise stated. We have

$$\frac{d^2 \lambda_\ell(r)}{dr^2} + \left\{ k^2 - \frac{\ell(\ell+1)}{r^2} \right\} \lambda_\ell(r) = U(r) \lambda_\ell(r) \quad (11)$$

with the following asymptotic form.

$$\lambda_\ell(r)_{r \rightarrow \infty} \rightarrow A_\ell \sin\left(kr - \frac{\ell}{2}\pi + \eta_\ell\right). \quad (12)$$

By employing the variable-phase approach(VPA) [21], we can directly solve η_ℓ as a function of r . We have

$$\frac{d\eta_\ell(r)}{dr} = -\frac{1}{k} U(r) G_\ell^2(\eta_\ell(r), kr) \quad (13)$$

with

$$G_\ell(\eta_\ell(r), kr) \equiv \cos(\eta_\ell(r)) \hat{j}_\ell(kr) - \sin(\eta_\ell(r)) \hat{n}_\ell(kr), \quad (14)$$

where $\hat{j}_\ell(kr)$ and $\hat{n}_\ell(kr)$ are, respectively, Ricatti-Bessel function of the first and second kind. We solve Eq. (13) with the boundary condition $\eta_\ell(0) = 0$; the scattering phaseshift η_ℓ is given by $\eta_\ell(r)$ at $r \rightarrow \infty$. We used the program VPA by Palov and Balint-Kurti [21] from the CPC Program Library for numerical calculations.

To investigate the resonance feature of $\ell = \ell_r = 4$ wave, we calculate η_ℓ for $\ell = 0-6$ and for the range of $E = 2.0 - 6.0 \text{ cm}^{-1}$. And, further, we confirmed that η_ℓ is negligibly small for $\ell > 6$ in the range of E considered.

Using the result of calculation, we evaluated the angle differential cross section $d\sigma/d\Omega$ for the range of polar angle $\theta = 0^\circ$ to 180° . Also, because we are considering the axial symmetric case, we evaluated the polar-angle differential cross section (polar-angle DCS) $d\sigma/d\theta$, which is given by

$$\frac{d\sigma}{d\theta} = 2\pi \sin \theta \cdot \left(\frac{d\sigma}{d\Omega} \right). \quad (15)$$

In Fig. 2, we show the energy and angle dependence of $(d\sigma/d\theta)$ for the range of $E = 2.0$ to 6.0 cm^{-1} and $\theta = 0^\circ$ to 180° . By plotting the polar-angle DCS, we can give a better appearance over the whole range of θ except the forward and backward extremes. We can clearly observe the presence of resonance around $E = 4.13 \text{ cm}^{-1}$. The asymmetry of the resonance profile varies with the change of θ due to the angle-dependent strength of the interference from the non-resonant partial waves, which will be discussed in detail later in this section. The lower entry of the figure gives the cross-section by Blatt and Biedenharn's formula [19] given in Eq. (1). The upper entry represents the synthesized cross section by the

polar-angle DCS

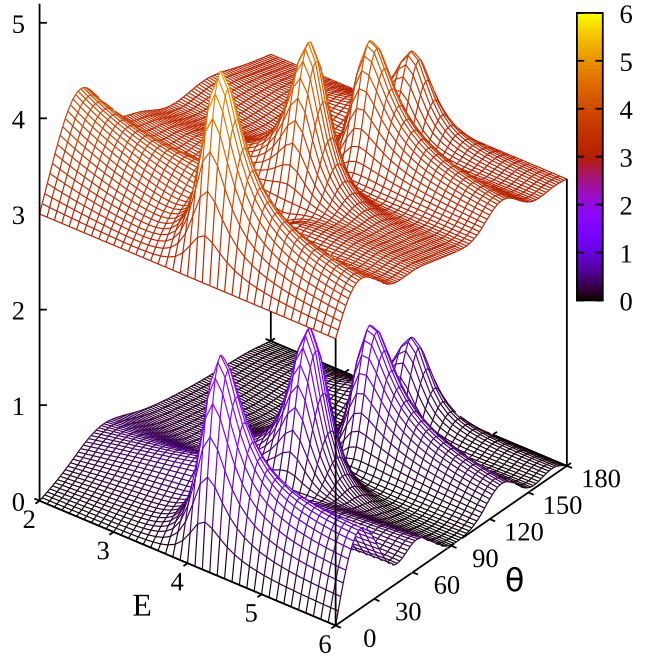


Fig. 2 The energy dependence of the polar-angle DCS $d\sigma/d\theta$ defined in Eq. (15). The lower surface: the cross section obtained from Blatt and Biedenharn's formula [19] given in Eq. (1). The upper surface: synthesized cross section given in Eq. (9). The values have been raised up by $3 \times 10^3 a_0^2$ to give a separate view of the cross sections. On the horizontal plane, E: center-of-mass collision energy in units of cm^{-1} , and θ : center-of-mass scattering angle in units of degree. Vertical axis: polar-angle DCS in units of $1 \times 10^3 a_0^2$

formula in Eq. (9). The values are raised up by 3000 a_0^2 to give a separate view of the cross sections. We can point out that Eq. (9) simulates well the rigorous cross sections by Blatt and Biedenharn [19].

To further investigate the effect of angle-dependent interference on the resonance profile, we analyze the behavior of the partial wave phaseshift η_ℓ in detail. In Fig. 3, we show η_ℓ for $\ell = 0$ to 6. We can observe a sharp resonance structure at $\ell = \ell_r = 4$, which was pointed out as an orbiting resonance by Toennies [7]. We performed a resonance phaseshift analysis to the ℓ_r partial wave. We fitted the resonance formula $\eta_{\ell_r} = \delta_{\ell_r} + \Delta_{\ell_r} = \delta_{\ell_r} - \cot^{-1}((E - E_r)/(\Gamma/2))$ given in Eq. (4) to the VPA-calculation. We found that the set of $\delta_{\ell_r} = -0.128$ radian, $E_r = 4.13 \text{ cm}^{-1}$, and $\Gamma = 0.56 \text{ cm}^{-1}$ gives a good fit to the calculated η_{ℓ_r} , which is given by a solid (red online) curve in Fig. 3. The fitted resonance curve is plotted by open (red online) circles in the figure. At the resonance point $E_r = 4.13 \text{ cm}^{-1}$, we evaluated the background phaseshifts δ_ℓ for $\ell = 0$ to 6. They are listed in Table 2 with the obtained resonance parameters. Although δ_ℓ inherently depends on the scattering energy E , we take their values at $E = E_r$ assuming that they are constant within the resonance width. This allows us to treat $w_{\ell_r}(\theta)$ as a function solely of θ . The overall agreement between the results of Eqs.(1) and (9) shown in Fig. 2 supports this viewpoint.

Using the values given in Table 2, we evaluate $w_{\ell_r}(\theta)$ in Eq. (6) and then $q(\theta)$ in Eq. (8). We used a program ANGMOM (catalogue number: ACYK) in CPC Program Library for Clebsch-Gordan coefficients [22]. In Fig. 4, we show the steric view of the θ evolution of $w_{\ell_r}(\theta)$ for $\ell_r = 4$ resonance at $E_r = 4.13 \text{ cm}^{-1}$ on a Gauss plane by solid

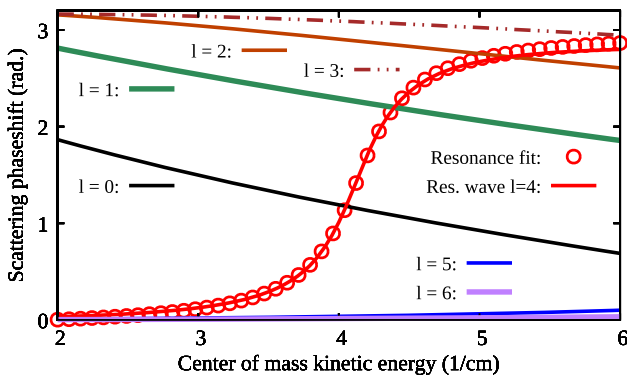


Fig. 3 Partial wave phaseshifts η_ℓ for $\ell = 0$ to 6 and resonance phaseshift analysis for $\ell_r = 4$. Dashed (black online) curve: $\ell = 0$, Dotted (sea-green online) curve: $\ell = 1$, Dot dashed (dark-orange online) curve: $\ell = 2$, Double dot dashed (brown online) curve: $\ell = 3$, Solid (red online) curve: $\ell = 4$, Dashed (magenta online) curve: $\ell = 5$, Dotted (purple online) curve: $\ell = 6$. Open (red online) circles show the fitting of the resonance formula $\eta_{\ell_r} = \delta_{\ell_r} + \Delta_{\ell_r} = \delta_{\ell_r} - \cot^{-1}((E - E_r)/(\Gamma/2))$ to the VPA-calculation for $\ell = 4$ partial wave

Table 2 Background phaseshifts δ_ℓ for $\ell = 0 - 6$ at $E = E_r = 4.13 \text{ cm}^{-1}$, and the resonance parameters E_r and Γ for $\ell_r = 4$ resonant partial wave

ℓ	δ_ℓ (radian)	E_r (cm^{-1})	Γ (cm^{-1})
0	1.153	4.13	0.56
1	2.258		
2	2.884		
3	3.084		
4	-0.128		
5	0.040		
6	0.015		

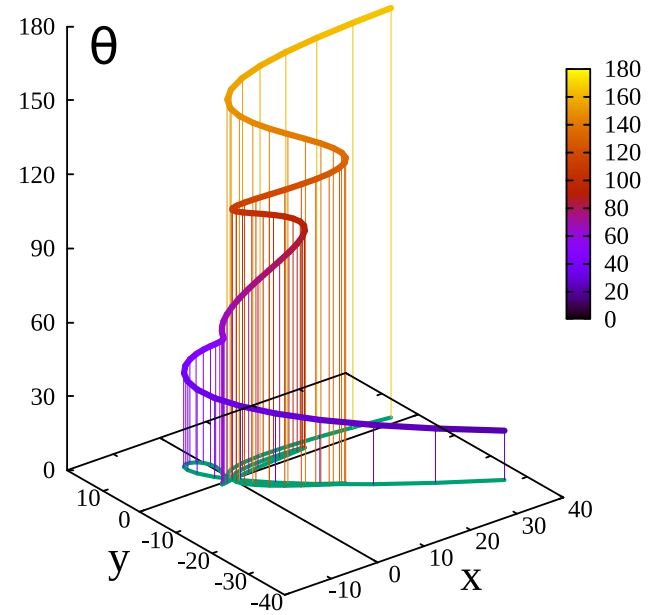


Fig. 4 Three dimensional scattering angle dependent evolution of the complex-number parameter $w_{\ell_r}(\theta)$ (See Eq. (6)). The set of horizontal axes x and y represents the Gauss plane $x + iy$. The vertical axis gives the scattering angle θ in units of degree. Thick solid curve with (color online) gradation: $w_{\ell_r}(\theta)$. The gradation(color online) illustrates the evolution of θ . Solid (green online) curve: Projection of $w_{\ell_r}(\theta)$ onto the Gauss plane

curve with (color online) gradation. The set of horizontal axes x and y represents the Gauss plane $x + iy$. The vertical axis gives the scattering angle θ in units of degree. We also give the projection of $w_{\ell_r}(\theta)$ onto the Gauss plane by solid (green online) curve. We find that $w_{\ell_r}(\theta)$ is continuous and smooth as a function of θ . We can thus visualize the interference feature of the shape and magnitude for the resonance profile.

To obtain a deeper insight into the resonance profile at moderate scattering angles, we evaluate the resonance profiles at several representative θ . In Fig. 5, we show the trace of $w_{\ell_r}(\theta)$ for $\theta = 23^\circ - 164^\circ$. Because the resonance

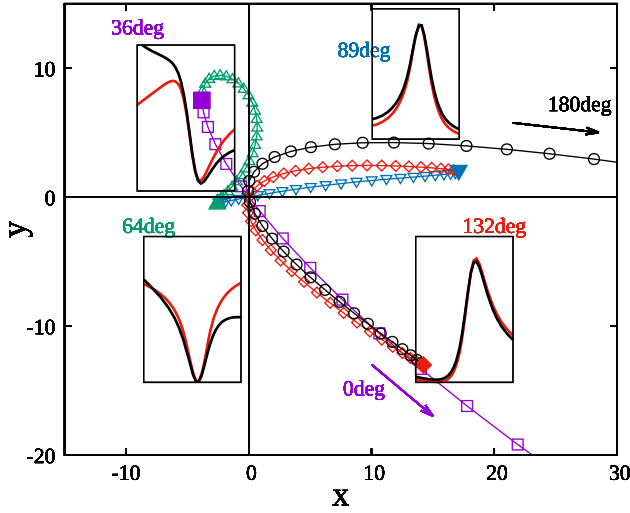


Fig. 5 The trace of $w_{\ell_r}(\theta)$ for $\theta = 20^\circ$ to 160° on the Gauss plane $x + iy$. Open square (magenta online): $\theta \leq 36^\circ$, Open triangle (green online): $36^\circ < \theta \leq 64^\circ$, Open inverse triangle (light blue online): $64^\circ < \theta \leq 89^\circ$, Open rhombus (red online): $89^\circ < \theta \leq 132^\circ$, and, Open circle (black online): $132^\circ < \theta$. Marks of the representative scattering angles are solid (magenta online) square on 36° , solid (green online) triangle on 64° , solid (light blue online) inverted triangle on 89° , and solid (red online) rhombus for 132° . The insets show the Fano resonance profiles at those angles. Solid curve: Eq. (9), and Dotted curve: Eq. (1)

is on $\ell_r = 4$ and $|w_{\ell_r}(\theta)|$ is its height, $w_{\ell_r}(\theta)$ vanishes 4 times through the whole range of θ , as discussed later in detail in the next section. In this figure, to visualize the angle evolution of $w_{\ell_r}(\theta)$ clearly, we divided the range of angles into several parts and illustrated the curve with different symbols and colors. For several representative scattering angles, which are marked on the $w_{\ell_r}(\theta)$ curve by solid (magenta online) square for 36° , solid (green online) triangle for 64° , solid (light blue online) inverted triangle for 89° , and solid (red online) rhombus for 132° , the corresponding normalized Fano profiles are given in the insets by (red online) solid curves (Eq. (9)). For comparison, we also give the correspondent angle differential cross-section (Eq. (1)) by (magenta online) dots. The range of E is from 3.0 to 5.0 cm^{-1} .

4. Results and Discussion

Fano resonance is known as one of the ubiquitous phenomena that are observed in isolated resonators that are embedded in broadband oscillations regardless of quantum or classical systems [2]. The Fano resonance in cold atomic collisions gives deep insight into the relation of quantum interference with inter-atomic interaction potential. Resonances in the elastic scattering of quantum mechanical particles can usually be observed when the de Broglie

wavelength is comparable to the range of interaction potential. For cold atomic collisions, their de Broglie wavelengths are long enough to satisfy such a condition. Because the resonance behavior is highly sensitive to the inter-atomic interaction potentials, the analysis of resonance profiles can provide a demanding test for potential evaluations.

In the present paper, we have pointed out that the angle-dependent evolution of the resonance profiles for cold atomic and molecular collisions can be characterized by a complex-valued parameter w_{ℓ_r} given in Eq. (6), following the discussion by Koike [6] on the resonance in slow electron scattering from atoms. The height of the resonance is given by the absolute value of $w_{\ell_r}(\theta)$, as seen in Eq. (9). We also find from Eqs.(8 and 9) that the angle-dependent asymmetry of the resonance profile can be characterized by the argument of the complex-number $w_{\ell_r}(\theta)$; Fano's asymmetry parameter q is expressed in terms of $\arg w_{\ell_r}(\theta)$ as in Eq. (8). The angle-dependent asymmetry of the resonance can be described by a single Fano profile function $(q + \epsilon)^2 / (1 + \epsilon^2)$ with an angle-dependent q -parameter $q(\theta)$.

We can calculate $q(\theta)$ using Eq. (8). In Fig. 6, we plot $q(\theta)$ for $\theta = 0^\circ$ to 180° . As found in this plot, $q(\theta)$ becomes discontinuous as a function of θ at several values of θ . These discontinuities in the q -parameter as a function of θ are related to the zero points of the θ -dependent partial wave functions $P_{\ell_r}(\cos \theta)$ of the resonant partial waves. For example, for $\ell_r = 4$, $w_{\ell_r}(\theta) = 0$ at $\theta = 30.56^\circ$ (see Fig. 5), and as θ increases and crosses the origin of the Gauss plane, $w_{\ell_r}(\theta)$ moves from quadrant 4, where $q > 0$, to quadrant 2, where $q < 0$. Thus, the q -parameter changes

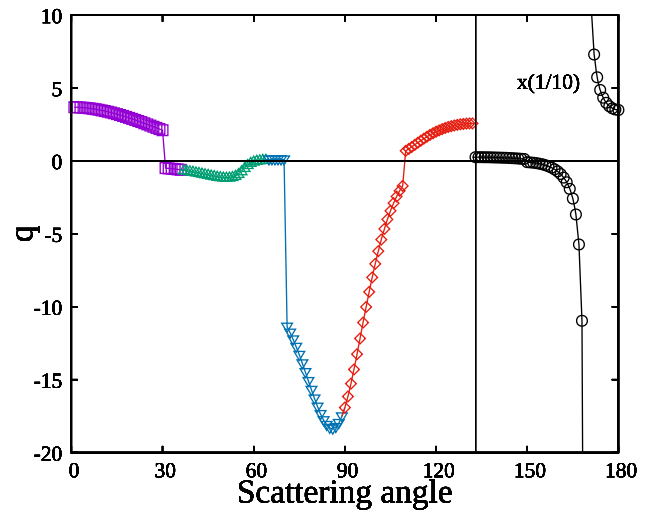


Fig. 6 Angle dependent Fano's q -parameter $q(\theta)$ for $\ell_r = 4$ shape resonance of $\text{H} + \text{Kr}$ elastic scattering. Horizontal axis: scattering angle θ in units of degree. Vertical axis: $q(\theta)$. For the symbols on the curve, see the caption in Fig. 5

discontinuously at the point $w_{\ell_r}(\theta) = 0$. We have to point out that this feature could cause difficulties in the resonance profile analysis of experimental spectra. Because the experiments must contain the data from a certain range of θ due to the resolution of the apparatus, it might sometimes be inappropriate to assign a single value of q for an experimental resonance profile. In contrast, as confirmed in Eq. (6), Figs. 4, and 5, the parameter $w_{\ell_r}(\theta)$ is continuous and smooth throughout the whole range of θ , which may ensure the stability of the spectral analysis.

We theoretically investigated the profile asymmetry of $\ell_r = 4$ orbiting resonance for H + Kr elastic scattering around $E_r = 4.13 \text{ cm}^{-1}$. As found in Figs. 4 and 5, $w_{\ell_r}(\theta)$ makes five wings in the scattering angles reflected by the character of $\ell_r = 4$ resonance wave, of which angular part is given by $Y_{\ell_r,0}(\theta, \phi) = \left(\frac{2\ell_r+1}{4\pi}\right)^{1/2} P_{\ell_r}(\cos \theta)$, where Y and ϕ are the spherical harmonics and the azimuth angle with respect to the incident beam direction, respectively. At the scattering angles that satisfies $P_{\ell_r}(\cos \theta) = 0$, the contribution of the ℓ_r resonant partial wave to the differential cross section vanishes and consequently the height of the resonance profile $|w_{\ell_r}(\theta)|$ becomes zero regardless of the presence of the other non-resonant partial waves. This can be clearly verified by examining the alternative expression of $w_{\ell_r}(\theta)$ given in Eq. (A.12). As realized in Eq. (9), $|w_{\ell_r}(\theta)|$ is the height of resonance, i.e., the distance between the top and bottom of the resonance profile, which is controlled solely by the geometry-dependent factors. In contrast, $\arg(w_{\ell_r}(\theta))$ determines the shape of the resonance profile, which is controlled solely by the effect of interference between the resonant and non-resonant partial waves. As seen in Fig. 5, $w_{\ell_r}(\theta)$ becomes zero for 4 times at the zero points of $P_4(\cos \theta)$, $\theta = 30.56^\circ, 70.12^\circ, 109.88^\circ$, and 149.44° , because the resonance height is zero when the amplitude of resonant partial wave is zero. At $w_{\ell_r}(\theta) = 0$, the resonance profile vanishes and the parameter $q(\theta)$ experiences a discontinuous change (see Fig. 6) at its passage of zero point. For example, at $\theta = 31^\circ$, $w_{\ell_r}(\theta)$ passes through the coordinate origin from the 4th quadrant to the 2nd quadrant on the Gauss plane with the increase of θ . The value of $q(\theta)$ jumps from 2.4 to -0.4 discontinuously, whereas $w_{\ell_r}(\theta)$ is continuous and smooth at the passage. Furthermore, we have to point out that $q(\theta)$ is divergent when $w_{\ell_r}(\theta)$ is real and positive. On the positive real axis of the Gauss plane, $\arg(w_{\ell_r}(\theta))$ is 0 (or 2π), and $q(\theta) \rightarrow \pm\infty$, accordingly. For example, we observe the divergence of q at $\theta = 169^\circ$ as seen in Fig. 6. In fact, $w_{\ell_r}(169^\circ) = 46.91 + 0.1875i$ and $w_{\ell_r}(170^\circ) = 50.47 - 0.4131i$. The parameter $w_{\ell_r}(\theta)$ crosses the positive real axis in between, and, therefore, $q(\theta)$ diverges between $\theta = 169^\circ$ and 170° . However, there are no physical significance for this divergence, because

both $q = \pm\infty$ indicate the same symmetric Lorentzian profile. In contrast, the parameter $w_{\ell_r}(\theta)$ has no such singularity of the behavior along the evolution of θ . This suggests that the use of $w_{\ell_r}(\theta)$ is advantageous for the analysis of experimental resonance profiles compared to following the θ dependence of $q(\theta)$.

We can further discuss the behavior of the angle dependent Fano's q -parameter in relation to $w_{\ell_r}(\theta)$. The parameter $q(\theta)$ is defined by Eq. (8). In the case $w_{\ell_r}(\theta)$ is in the 1st or 2nd quadrant on the Gauss plane, in other words, imaginary part of $w_{\ell_r}(\theta)$ is positive, $q(\theta)$ is negative and the resonance profile has a minimum below the resonance point energy E_r due to the destructive interference with the background scattering waves. On the other hand, in the case $w_{\ell_r}(\theta)$ is in the 3rd or 4th quadrant, $q(\theta)$ is positive and the minimum is above E_r . Now, we consider the change of $q(\theta)$ at the node of the resonant partial wave, namely, at $\theta = \theta_0$ that satisfies $P_{\ell_r}(\cos \theta) = 0$. As seen in Eq. (6) or Eq. (A.12), $w_{\ell_r}(\theta)$ tends to zero when θ tends to θ_0 , whereas $\arg(w_{\ell_r}(\theta))$ remains finite and, consequently, $q(\theta)$ remains finite. However, the sign of $w_{\ell_r}(\theta)$ is expected to be different between the region $\theta < \theta_0$ and $\theta > \theta_0$ due to the terms of interference with the other non-resonant partial waves. At $\theta = \theta_0$, $P_{\ell_r}(\cos \theta)$ changes sign and the constructive interference turns to destructive interference, vice versa. Consequently, $q(\theta)$ experiences a discontinuous change at the zero point of $P_{\ell_r}(\cos \theta)$.

Paliwal et al. [17] have carried out a challenging measurement on the Fano interference in angle-resolved elastic $\text{He}^* + \text{D}_2$ scattering for the $\ell_r = 6$ partial wave at the resonance point $E_r = 4.8$ Kelvin. They found the angle-dependent evolution of the asymmetry in the resonance profile. They assigned it as the Fano resonance. They opened a wide area of research field in cold atomic resonance scattering. They proposed, however, a profile formula that does not meet the conventional understandings for the value of q -parameter. Fano's profile function $(q + \epsilon)^2 / (1 + \epsilon^2)$ gives a window-type profile at $q = 0$, whereas their formula gives Lorentzian instead. It is, however, desirable to maintain the current convention on the values of q for the assignment of the shape of resonance. As shown in Eq. (9) of the present paper, the resonance profile can be represented rigorously by the conventional shape of Fano profile formula. It would be advantageous for further development of the resonance study in cold collisions to use a set of formulae that are analytic and maintain connectivity with the conventional theory of Fano resonance [1]. And, furthermore, we note that $q(\theta)$ of their resonance may change its sign for 6 times or more during the whole passage of θ from 0° to 180° , because they identified the resonance as of $\ell_r = 6$. The use of $w_{\ell_r}(\theta)$ in the spectral

analysis may give an advantage for overcoming the limited angle resolution of the experimental apparatus.

5. Conclusion

For angle-differential elastic processes in cold atomic collisions, the resonance line shape has been rigorously shown to be given by Fano's profile formula $(q + \epsilon)^2 / (1 + \epsilon^2)$ with an angle-dependent Fano's profile parameter $q = q(\theta)$. Explicit expression of $q(\theta)$ has been derived by introducing a complex number parameter w_{ℓ_r} which is a smooth function of θ on the Gauss plane. The resonance parameters $q(\theta)$ and $w_{\ell_r}(\theta)$ were derived in analogy to a previous study of slow electron-atom collisions, where the de Broglie wavelength of the collision system is similar to that of cold atomic and molecular collisions. Our paper presents new possibilities for our theoretical formalism, going beyond the previous electron-atom collision study to angle-dependent resonance profile analysis in the newly developing field of cold atomic and molecular collisions.

The Fano resonances in H + Kr differential elastic cold collisions have been theoretically studied using the potential given by Toennies et al. [7]. The resonance parameters $q(\theta)$ and $w_{\ell_r}(\theta)$ are quite sensitive to the form of interaction potentials, providing us with an opportunity for effective tests for the potential evaluation.

Appendix: Detailed derivation of Eq. (6)

The angle-dependent parameter $w_{\ell_r}(\theta)$ defined in Eq. (6) can be introduced by considering the resonance characteristics in Eqs. (1) and (2) in the following way. We assume an isolated resonance with resonance and background phaseshifts Δ_{ℓ_r} and $\delta_{\ell} (= \delta_{\ell_r})$ for $\ell = \ell_r$ as given in Eq. (4). The phaseshift part of B_L of Eq. (2) $\sin \eta_{\ell} \sin \eta_{\ell'} \cos(\eta_{\ell} - \eta_{\ell'})$ can be reduced as the linear combination of $\cos(2\Delta_{\ell_r})$, $\sin(2\Delta_{\ell_r})$, and a background term. We first consider the case of $\ell' = \ell_r$ and $\ell \neq \ell_r$. In this case, $\eta_{\ell'} = \delta_{\ell_r} + \Delta_{\ell_r}$ and $\eta_{\ell} = \delta_{\ell}$, and we obtain

$$\begin{aligned} \sin \eta_{\ell} \sin \eta_{\ell'} \cos(\eta_{\ell} - \eta_{\ell'}) &= \sin \delta_{\ell} \sin(\delta_{\ell_r} + \Delta_{\ell_r}) \cos(\delta_{\ell} - \delta_{\ell_r} - \Delta_{\ell_r}) \\ &= \frac{1}{2} \sin \delta_{\ell} \sin(2\delta_{\ell_r} - \delta_{\ell}) \cos(2\Delta_{\ell_r}) \\ &\quad + \frac{1}{2} \sin \delta_{\ell} \cos(2\delta_{\ell_r} - \delta_{\ell}) \sin(2\Delta_{\ell_r}) \\ &\quad + \frac{1}{2} \sin^2 \delta_{\ell}. \end{aligned} \quad (\text{A.1})$$

We can also consider the other cases such as $\ell = \ell' = \ell_r$ or $\ell, \ell' \neq \ell_r$. Substituting all the results into Eq. (2), we obtain the following form for B_L .

$$B_L = -\frac{1}{2} X_L \cos(2\Delta_{\ell_r}) + \frac{1}{2} Y_L \sin(2\Delta_{\ell_r}) + \frac{1}{2} Z_L \quad (\text{A.2})$$

with

$$\begin{aligned} X_L &= (2\ell_r + 1)^2 [C(\ell_r \ell_r L; 00)]^2 \cos(2\delta_{\ell_r}) \\ &\quad - 2 \sum_{\ell=0, \ell \neq \ell_r}^{\infty} (2\ell + 1)(2\ell_r + 1) [C(\ell \ell_r L; 00)]^2 \end{aligned} \quad (\text{A.3})$$

$$\times \sin \delta_{\ell} \sin(2\delta_{\ell_r} - \delta_{\ell}),$$

$$\begin{aligned} Y_L &= (2\ell_r + 1)^2 [C(\ell_r \ell_r L; 00)]^2 \sin(2\delta_{\ell_r}) \\ &\quad + 2 \sum_{\ell=0, \ell \neq \ell_r}^{\infty} (2\ell + 1)(2\ell_r + 1) [C(\ell \ell_r L; 00)]^2 \end{aligned} \quad (\text{A.4})$$

$$\times \sin \delta_{\ell} \cos(2\delta_{\ell_r} - \delta_{\ell}),$$

$$\begin{aligned} Z_L &= (2\ell_r + 1)^2 [C(\ell_r \ell_r L; 00)]^2 \\ &\quad + 2 \sum_{\ell=0, \ell \neq \ell_r}^{\infty} (2\ell + 1)(2\ell_r + 1) [C(\ell \ell_r L; 00)]^2 \sin^2 \delta_{\ell} \\ &\quad + 2 \sum_{\ell=0, \ell \neq \ell_r}^{\infty} \sum_{\ell'=0, \ell' \neq \ell_r}^{\infty} (2\ell + 1)(2\ell' + 1) [C(\ell \ell' L; 00)]^2 \\ &\quad \times \sin \delta_{\ell} \sin \delta_{\ell'} \cos(\delta_{\ell} - \delta_{\ell'}). \end{aligned} \quad (\text{A.5})$$

We substitute eq.(A.2) into Eq. (1), obtaining the following form for the differential cross section.

$$\frac{d\sigma}{d\Omega} = \frac{1}{k^2} \left\{ -\frac{1}{2} x \cos(2\Delta_{\ell_r}) + \frac{1}{2} y \sin(2\Delta_{\ell_r}) + \frac{1}{2} z \right\} \quad (\text{A.6})$$

with

$$x = \sum_{L=0}^{\infty} X_L P_L(\cos \theta), \quad (\text{A.7})$$

$$y = \sum_{L=0}^{\infty} Y_L P_L(\cos \theta), \quad (\text{A.8})$$

$$z = \sum_{L=0}^{\infty} Z_L P_L(\cos \theta). \quad (\text{A.9})$$

We can see that the differential cross section is represented by the linear combination of $\cos(2\Delta_{\ell_r})$ and $\sin(2\Delta_{\ell_r})$. Recall that Δ_{ℓ_r} varies from 0 to π as the scattering energy increases at the resonance point $\Delta_{\ell_r} = \pi/2$; $\cos(2\Delta_{\ell_r})$ represents the symmetric part of the resonance and $\sin(2\Delta_{\ell_r})$ represents the anti-symmetric part of the resonance. The ratio of x and y then gives the asymmetry of the resonance profile. Meanwhile, $\sqrt{x^2 + y^2}$ gives the resonance height. As found in Eqs. (A.7) and (A.8), x and y are functions of the scattering angle θ . We can trace the change of the resonance profile in θ through the change of x and y .

To illustrate those resonance characteristics in terms of a single variable, it is plausible to introduce a complex-

valued expression. We introduce a parameter $w_{\ell_r}(\theta)$ in the following form.

$$w_{\ell_r}(\theta) = x + iy \quad (\text{A.10})$$

Using Eqs. (A.7)–(A.9) and (A.3)–(A.5), we can derive an explicit expression for $w_{\ell_r}(\theta)$. After a straightforward calculation, we obtain $w_{\ell_r}(\theta)$ introduced in Eq. (6).

We can further obtain a different representation for $w_{\ell_r}(\theta)$ that gives clearer insight for the interference effect in the resonance profile. We recall the following connection rule of the Legendre polynomial [20].

$$P_{\ell}(\cos \theta) P_{\ell'}(\cos \theta) = \sum_{L=0}^{\infty} [C(\ell \ell' L; 00)]^2 P_L(\cos \theta). \quad (\text{A.11})$$

Taking the L -sum first in Eq. (6), we can reduce $w_{\ell_r}(\theta)$ in the following form.

$$\begin{aligned} w_{\ell_r}(\theta) = & (2\ell_r + 1) \exp(2i\delta_{\ell_r}) \{ (2\ell_r + 1) P_{\ell_r}(\cos \theta) \\ & + \sum_{\ell=0, \ell \neq \ell_r}^{\infty} (2\ell + 1) [1 - \exp(-2i\delta_{\ell})] P_{\ell}(\cos \theta) \} \\ & \times P_{\ell_r}(\cos \theta). \end{aligned} \quad (\text{A.12})$$

We can see that the first term in the brace $\{ \}$ gives the asymmetry property caused by the background scattering of the resonant partial wave, and that the latter terms give the effect of interference from the other non-resonant partial waves. Due to the presence of the latter terms, the argument of complex value $w_{\ell_r}(\theta)$, which is directly connected to the q -parameter as in Eq. (8), may be different for different scattering angles.

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